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Innovation Processes in Higher Education

Summary

The paper contains a brief description of the research we have undertaken on innovations and innovation processes in higher education [1]. Within the framework of a larger project (a study of educational planning for Swedish Higher Education [2]), we have made seven case studies of innovation processes and tried to relate them to an attempt towards a general theory of innovations. We have built on, and tried to exemplify, the field theory of the social psychologist, Kurt Lewin [3]. This means that we view change as being realised through a form of political process, determined by the strength of the different forces within a social system. We have tried to define the characteristics of higher education as a social system and the factors which determine the nature of the various forces towards a certain type of change.

The Confused State of Innovation Research

Research about innovations in education should be assessed as part of innovation research in general. In the optimistic climate of the 1950s and 1960s it was thought that research would soon clarify the conditions of innovation diffusion and social change, and that this would help mankind to change more rapidly and, consequently, be more happy.

In the 1970s, things have come to seem more complicated. Innovation researchers have begun to feel disappointment, insecurity, and almost irritation:

Unfortunately, the theoretical value of the research that has been done is ambiguous. Perhaps the most alarming characteristic of the body of empirical study of innovation is the extreme variance among its findings, what we call instability. Factors found to be important for innovation in one study are found to be considerably less important, not important at all or even inversely important in another study. [4]

Our explanation of this phenomenon is twofold: first, that due regard has not been paid to the differences between different types of innovations; and secondly, that due regard has not been paid to the differences between the social systems within which the innovation processes take place.

It may be added that the difficulties of achieving a comprehensive and holistic account of innovations derive from the fragmentation of discipline-based theories, which illuminate social reality from the point of view of a particular discipline without attempting a more general overview. What is called 'a systems approach' represents one attempt to escape this limitation.

The current situation in innovation research is also related to what seems to us a general methodological crisis in the social sciences. The belief is no longer valid that the methods of the natural sciences offer the only acceptable approach. It is increasingly questioned whether relevant explanations of social realities can be found by defining isolated variables in social events and relating them to each other in pairs [5].

Diffusion Models and Political Models

The classical tradition in innovation research regards the development of innovations as a diffusion process. According to this tradition, diffusion occurs between individuals, even though the innovations take place in a geographical and social field, the structure of which influences the diffusion. Diffusion is seen as dependent on geographical communications, on information through social contacts, and individual acceptance occurs through imitation or through some process of intellectual conviction. Research of this kind is rooted in ethnological research on innovation diffusion in ancient times and within primitive cultures. It was carried over to research on the diffusion of industrial products in modern societies.

But, in parallel with diffusion research, there has grown up a research field oriented towards change not merely *within* a system but also *of* a social system. In this setting, there is a greater emphasis on the behaviour of groups and organisations and on interaction within and between groups. The main intellectual debt is to the ideas of social anthropology, to social psychology and sociology, and to the type of organisation research that began to develop around the problems encountered by industrial enterprises. This approach, to which we ourselves admit, can be described as a combination of a systems approach and a contingency approach.

There is now a growing tendency to distinguish between various types of innovations, related to different models of the innovation process. R. Danforth Ross has differentiated between paradigmatic and extraparadigmatic innovation [6]. The types of innovations we have studied are of the latter type. We prefer another terminology however—system-consistent and system-divergent innovations—as being more in accordance with our framework of analysis.

System Characteristics within Higher Education

Our terminology is designed to emphasise that the innovation process is either consistent with or divergent from the main characteristics of the system. In the former case the process is one of dissemination, and in the latter of a political battle.

What, then, are the characteristics of the system which determine its reactions to a certain type of innovation? First and foremost, in our study, there are certain national characteristics of the Swedish system of higher education. The most relevant of them is the comparatively high degree of central (national) power and authority. We would argue, however, that the basic properties of higher education as a social system are largely the same in most countries.

We claim that all social systems can be described and analysed by means of five main properties: membership composition, ideology, technology, organisational structure, and relations with the environment. As far as higher education is concerned, some brief comments can usefully be made on each.

The most important membership group is that of the teachers and researchers, who divide themselves into various types of overlapping sub-systems. Administrators and students are groups which have a growing importance. The specific ideology, i.e. goal perceptions and values, is ultimately determined by the disciplines which teachers and researchers profess. Within each discipline, graduate education, research and undergraduate education (usually in this order of priority) are dominant goals. The specific technology of higher education and research renders innovations highly sensitive to personal behaviour and social interaction. The disciplinary orientation is clearly reflected in the organisational structure. Finally, in their relations with the environment, higher education institutions in most countries have an important common denominator: they are not exposed to economic competition and are not under market pressure to innovate in order to survive.

Background and Purpose of our Study

Our study of innovations and innovation processes is part of a functional study of the Swedish Higher Education system, which in its entirety can be characterised as a clinical examination of governance, planning systems and innovativeness in higher education.

Resting in balance, any system appears static and formalised—as in an organisation chart, in a law text or in an official investigation report. But when a disturbance is introduced into the system—and an innovation is frequently a disturbance—a dynamic force field emerges; life and reality enters the system. One can observe interactions between formal and informal subsystems, interactions which bear no correlation with the above-mentioned world of organisation charts, law texts or official investigation reports. Just as there is a ‘hidden curriculum’ in education, there is a ‘hidden governance system’, a ‘private life of higher education’ which can be traced under the superstructure of formal organisations and regulations.

By looking closely into some cases of innovation processes and analysing them in terms of a uniform and consistent model, we feel we have obtained a more thorough knowledge of the private life of higher education in Sweden and deepened our understanding of the conditions for change in the academic world.

Methodological Aspects

Our study encompassed seven cases of innovation processes, all of which concerned innovations of contents or methods and may be classified as systemdivergent innovations. The cases were all selected on subjective grounds—that is, we selected innovations of which we already knew. But we are confident that we could have selected any innovation of a systemdivergent kind and still have come to the same conclusions.

There are at least three techniques for studying an innovation process. One is continuously to observe the process as it unfolds. Another is to participate actively in order to influence the outcome. The third is to study the innovation retrospectively and try to reconstruct the process. All three methods have their pros and cons. In our case, as it happened, we had no choice, since our time frame eliminated the first two possibilities. One definite advantage of the one we were forced to adopt is that it did not in any way affect the innovation process itself.

Our data have consisted partly of written documents and partly of loosely structured interviews with key persons involved in the process. We would describe our method of investigation and analysis as an *illuminative method*, in which emphasis is put on illumination rather than on measurement; on observation and interpretation of the process in its entirety rather than on the collection and classification of data concerning certain selected variables.

Theoretical Basis

Our main theoretical starting point has been a systems approach oriented towards social psychology. In the development of our theoretical framework and conceptual apparatus we have leaned heavily on the German-American social psychologist Kurt Lewin (1890–1947). A systems approach implies that events and behaviours can be explained by means of structures and forces across the social field as a whole. The importance of various parts of the system does not follow from their inherent characteristics but from their positions within the system. A social system (characterised by its main properties—membership, ideology, technology, organisational structure and relations to the environment), is a grouping of people around certain purposes. Such a system has a natural tendency to preserve the interests of its members, its ideology, its technology and so on—that is, to resist change affecting *any* of its main properties.

In analysing our seven studies of innovation processes we have first made an attempt to appraise the situation as a whole, in accordance with the thinking of Lewin. We have then identified certain key features of the innovation process in terms of various dynamic constructs (a term borrowed from Lewin, even though the constructs themselves are not). Generally, our way of analysing change draws upon the idea of change as a disturbance of the so-called quasi-stationary equilibrium in a force field; the equilibrium is maintained by an equal distribution of forces on each side of the equilibrium; a disturbance in the force field is caused by driving forces gaining dominance over opposing forces.

When the equilibrium is disturbed, there is a change in the force field as a whole, which can be designated as an ‘unfreezing’. The next phase in the change process is ‘moving’: this continues until a new equilibrium is reached, which we refer to as ‘refreezing’. We have found these three phases between them to characterise the whole span of an innovation process.

When analysing an innovation process it is necessary to begin with a mapping of the field in its entirety, and of the various subsystems and their positions. In our interpretation of Lewin’s model, innovation becomes a political process in the sense that there are opposing forces fighting for dominance around the equilibrium.

We stated earlier that a system tends to preserve itself and to resist change. Nevertheless, we would argue that systemdivergent change is possible under certain conditions. We would also argue that most systems of higher education could and should develop a higher capacity for systemdivergent change.

Of course change can be forced upon a system—as a matter of fact, most national reforms within the Swedish system of education have been based upon the idea that change can be forced. A limited amount of innovation can undoubtedly be imposed by changes in rules and regulations. But when it comes to the types of change with which we have been concerned—namely relating to contents and methods of education—these coercive strategies do not seem to work, in large part (it seems) because to produce new knowledge or transmit it to students is itself a creative activity at the heart of the system and closely depending on its main properties. Such innovations cannot simply be inserted from outside: they have to be created anew within the system, by those who are members of it.

But if any system has a tendency to resist change, how is change possible? How can we understand that systemdivergent innovations do occur? The answer is that few systems are completely homogeneous. There may be—and this is a precondition for change—lack of concordance, for instance in membership interests or ideology. This lack of conformity gives rise to ‘cracks’ or conflicts in the system, and these in turn constitute a further precondition for change. Change also presupposes that the system is open, i.e.

that is has contacts with its environment. One could put it that such contacts allow impulses from the outside to flow into the 'cracks' in the system and to break the system by creating a potential for moving.

Starting from these general ideas on systemdivergent change, we have built up a conceptual apparatus which has enabled us to analyse the process of a variety of specific innovations within higher education. In building up this conceptual framework we have divided the constructs into two groups: descriptive categories for internal and external relations and analytic categories for the determining factors of the innovation process.

Descriptive Categories for Internal and External Relations

The innovation processes we have studied bear the characteristics of political processes. The concept of *level of conflict* becomes important in catching this aspect of the process. The level of conflict expresses, among other things, the perceptions of threat within systems and parts of systems during the various phases of an innovation process. Our case studies suggest that if a change of the state of equilibrium is caused mainly by an increase of the driving forces, there tends to be a high level of conflict, whereas a weakening of the restraining forces leads to a low level of conflict.

Visibility is closely related to the level of conflict. High visibility causes the opposition to notice system divergent or threatening characteristics of the innovation, and consequently to mobilise restraining forces. Low visibility can make it easier to implement an innovation—but a certain amount of visibility is necessary in order to rally support. A strategy of selective visibility may be the best way out of this dilemma.

Personnel changes relate to members entering or leaving the system or changing their positions within the system. In all the cases we studied there have been obvious personnel changes that have influenced the innovation process.

We already suggested that an innovation is always related to *external change*, although the external changes in question may vary in nature. Our seven case studies of innovation processes can only indicate the importance of external relations: it would be valuable to have another investigation which attempted a more complete mapping of those external changes contributing to changes in the higher education system.

Analytic Categories for Determining Factors

While we have some doubt about the value of our descriptive categories, we feel that our analytic categories do relate to fundamental issues. The literature on innovation abounds in designations of persons and roles deriving from the tradition which regards innovation as essentially a matter of diffusing information. We abandoned the resulting 'laundry lists' as quite unsuitable to our conception of innovation as a political process within a dynamic force field. The four concepts, or dynamic constructions, we have found most relevant to describe innovative activities are: gain/loss, ownership, leadership and power.

Gain/loss is the construct that best explains the reactions of subsystems and individuals to an innovation. It is a broad and multidimensional concept that covers a whole range of needs and interests. Initially we believed that security/stability would be the most obvious gain, and insecurity/instability the most obvious loss. This is possibly true if these concepts are seen in their widest frame of reference, including not only physical survival and security but also the security of status, of having one's position and one's recognised competence. Another aspect of gain/loss has more weight than we had originally expected, an aspect that our case studies have made us recognise as an obvious explanation of the intensity of certain goal-oriented behaviours. It is the question of ideological need, ideological gain or loss. From the standpoint of the system, the ideology of the system must be maintained. From an individual standpoint, the most valuable

gain is one of having created something that gives satisfaction. The best term from both standpoints may be self-realisation. It is impossible to weigh the needs of security and self-realisation against one another, but these two broad types of need, and the gains and losses associated with them, are what most noticeably place systems and individuals into the category of driving or restraining forces.

Ownership has emerged as a powerful explanation of attitudes towards innovation. It is possible to make a distinction between direct and indirect ownership. Total or direct ownership belongs to those who have created an innovation or introduced it in a complete or near-complete form into the system. Indirect ownership means that a person considers himself or herself to espouse a broad tendency to change which includes the innovation in question. This type of ownership is common among the formal leaders of the higher education system.

When it comes to *leadership*, we need to distinguish between innovation leaders, secondary leaders and formal leaders of innovative systems. The relationship between formal leadership and ownership has been remarked. The simple conceptual apparatus we have used to define leadership represents an effort to get away from the verbose and diffuse designation of individual roles which is common in innovation research. For our purposes, designations in terms of diffusion and information models—such as early and late adopters—are meaningless. For those cases, such as our own, where the innovation process is political in character, we find our conceptual apparatus sufficient.

However strong the motivation of a person or a group in terms of hopes of gain or risks of loss, however obvious the ownership or leadership of the innovation, if there is no *power* to be mobilised then there is no possibility of moving the equilibrium of the force field so that an innovation can be accomplished. The activation of power determines the development. However, non-activated power may influence the process as well. The mere knowledge that power does indeed exist in various parts of the system, and may be activated in a certain direction and under certain conditions, can itself cause other components of the system to increase or reduce the force they exercise, or even persuade them to change sides in the field.

Theoretical Conclusions

The main points in our attempt at formulating a theory of innovation would be as follows:

- (1) Any innovation theory must be limited to a certain type of innovation within a certain kind of system. Certain of its elements may be more generally valid, i.e. various innovation theories may overlap. We do not want to argue that our own attempted theory is valid for any system other than that of higher education. We regard it as valid only for innovations that are systemdivergent and affect contents and methods of education. Nevertheless, it may be the case that points (2)–(8) below are generally valid for systemdivergent innovations in other complex formal systems as well.
- (2) The conditions for change are determined mainly by systems characteristics.
- (3) The major characteristics of a system are membership composition, ideology, technology, organisational structure, and the nature of its relations with the environment.
- (4) Systems tend to defend the interests of their members, their ideology, their technology, their organisational structure, and the nature of their relations with the environment.

- (5) The innovation process is a political process which takes place in a field of mutually opposing forces around an equilibrium. The development, in accordance with Lewin's theory, needs to be explained in terms of the state of the field as a whole.
- (6) The main phases of the innovation process are unfreezing, moving, and refreezing. Unfreezing signifies that the possibility for change is created. Moving denotes a continuous disequilibrium, caused by the dominance of the driving forces over the restraining forces. Refreezing means that balance is created around a new equilibrium.
- (7) Innovations which deviate from the system can occur only under the following conditions:
 - (a) there must be a 'crack' in the system. The 'crack' is created when components of the system deviate from its main characteristics;
 - (b) there must be a connection between the 'crack' and the environment;
 - (c) impulses from the environment flow into the 'crack'.
- (8) In a specific system, the conditions for change are determined by the detailed contents of its main characteristics.
- (9) The total field situation can be defined by an analysis based on points (2)–(8) above; the field as a whole influences the general conditions for change and non-change. In order to refine the definition of change in a specific case, it is necessary to develop a conceptual apparatus that explains relevant interactions and interrelations in the field.
- (10) A tentative framework for describing aspects of the innovation process would consist of the concepts of level of conflict, visibility, personnel changes, and correlations with external changes.
- (11) The innovation process can be explained by means of the four concepts of gain/loss (needs, interests), ownership, leadership and power.

Final Comments

In discussing our investigation and our conclusions with members of the higher education system, we frequently meet with the question: do you regard change as something inherently good in itself? In innovation studies it is of course tempting to present the pro-innovation forces as positive and the anti-innovation forces as negative. However, innovation *per se* is neither good nor bad. It is an important task of the members of the higher education system to be able critically to examine new ideas and reject undesired change. There is, however, no denying that we generally look upon a high degree of innovativeness and capacity for change as desirable qualities. Bearing in mind the tendency of a system to preserve itself and to resist changes which may form a threat against its main characteristics, we believe that conscious efforts need to be made to counteract this tendency.

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